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NMAM INSTITUTE OF TECHNOLOGY, NITTE
Off-Campus Centre of Nitte (Deemed to be University)
Fourth Semester B. Tech (CBCS) Degree Examinations

May 2025

RI2109-1 – MICROCONTROLLER AND ITS APPLICATION
(For RI)

Duration: 3 Hours

Max. Marks: 100

Note:

Part – A: Multiple Choice Questions: Answer all **Twenty** questions in the **OMR Sheet** provided. Each question carries equal marks.

Part – B: Descriptive Answer Questions: Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II each** and **One full** question from **Unit – III**.

PART - A: MULTIPLE CHOICE QUESTIONS**20 Marks**

1. What is the size of the ALU in the 8051 microcontrollers?
A) 8 bytes
B) 8 bits
C) 16 bits
D) 16 bytes
2. How many bits are the registers in the 8051 microcontrollers?
A) 8 bits
B) 8 bytes
C) 16 bits
D) 16 bytes
3. Which of the following is a feature of the 8051 microcontrollers?
A) Single-cycle instruction execution
B) No internal RAM
C) On-chip oscillator
D) Does not support external memory
4. The 8051 microcontrollers can be used in communication systems for:
A) Only wired communication protocols
B) Only wireless communication protocols
C) Both wired and wireless communication protocols
D) Neither wired nor wireless communication protocols
5. What does the RETI instruction do in 8051 assembly language?
A) Re-enables all interrupts
B) Returns from a subroutine
C) Resets the microcontroller
D) Returns from an interrupt service routine
6. The DJNZ instruction is primarily used for:
A) Arithmetic operations
B) Looping operations
C) I/O operations
D) Direct memory access
7. Which addressing mode is commonly used for accessing constants in 8051?
A) Relative addressing
B) Direct addressing
C) Indexed addressing
D) Immediate addressing mode
8. What is the bit size of the 8051 microcontroller's accumulator?
A) 8 bits
B) 8 bytes
C) 16 bits
D) 16 bytes
9. Which port of the 8051 microcontroller is not bit-addressable?
A) Port 0
B) Port 2
C) Port 1
D) Port 3
10. Which of the following ports are multiplexed for address/data in an 8051 microcontroller?
A) Port 0
B) Port 2
C) Port 1
D) All ports are.

11. In which application is PWM (Pulse Width Modulation) most commonly used with the 8051?
 - A) Data encryption
 - B) Text display on LCD
 - C) Temperature measurement
 - D) Motor speed control
12. Which port of the 8051 microcontroller is also known for providing serial communication control signals?
 - A) Port 0
 - B) Port 2
 - C) Port 1
 - D) Port 3
13. Which MODE in 8051 is auto-reload mode?
 - A) MODE1
 - B) MODE2
 - C) MODE3
 - D) MODE0
14. Which MODE in 8051 can achieve a maximum delay?
 - A) MODE1
 - B) MODE2
 - C) MODE3
 - D) MODE0
15. Which application might require the use of external memory interfacing with the 8051?
 - A) Simple LED blinking
 - B) Storing large amounts of sensor data
 - C) Implementing a basic calculator
 - D) Basic temperature monitoring
16. To configure all pins of Port 1 as output on an 8051 microcontroller, which of the following instructions should be used?
 - A) MOV P1, #0x00
 - B) CLR P1
 - C) SETB P1
 - D) MOV P1, #0xFF
17. In ARM architecture, ARM stands for
 - A) Auto reload mode
 - B) Advanced RISC Machine
 - C) Auto Reset Mode
 - D) None of these
18. Which ARM instruction is used for conditional execution based on flags?
 - A) MOV
 - B) ADD
 - C) MOVX
 - D) CMP
19. Which mode is the ARM processor in immediately after reset?
 - A) USER MODE
 - B) SYSTEM MODE
 - C) SUPERVISOR MODE
 - D) IRQ MODE
20. What is the length of ARM instructions?
 - A) 32 bits
 - B) 8 bits
 - C) 16 bits
 - D) 64 bits

PART - B: DESCRIPTIVE ANSWER QUESTIONS

Unit – I

	Marks	BT*	CO*	PO*
1. a) With a neat architectural diagram explain the functions of each block in 8051 microcontrollers.	8	L*2	1	1
b) Explain PUSH and POP instructions in 8051 microcontrollers with an example for each case.	8	L2	2	1
2. a) With the neat diagram, explain 128 bytes RAM memory organization of 8051 microcontroller with address in detail.	8	L2	1	1
b) Explain the different addressing modes of 8051 with examples of each.	8	L3	2	1
3. a) Write a note on: Stack, Stack Pointer, Program counter, DPTR.	8	L2	1	1
b) Write a program to find the sum of all the numbers. Assume that 5 BCD data items are stored in RAM locations starting at 40H. Result should be in 50H.	8	L3	2	1

Unit – II

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|-------|--|---|----|---|---|
| 4. a) | Differentiate and explain I2C and SPI communication protocols. | 8 | L2 | 3 | 1 |
| b) | With an interfacing diagram & write an assembly language program to display number "HELP" on a multiplexed seven segment display I/O port pins. | 8 | L2 | 4 | 1 |
| 5. a) | In detail explain the basics of Interrupts in 8051 microcontrollers. Explain IE and IP registers also. | 8 | L2 | 3 | 1 |
| b) | Explain the importance of SCON register in detail in 8051 programming. | 8 | L2 | 4 | 1 |
| 6. a) | Write an assembly language program to generate a delay of 1 millisecond using Timer 0 in the 8051 microcontrollers. Assume a crystal frequency of 11.0592 MHz. | 8 | L3 | 3 | 1 |
| b) | Explain the different Modes of TIMER/COUNTER with block diagram. | 8 | L2 | 4 | 1 |

Unit – III

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|-------|--|---|----|---|---|
| 7. a) | How are barrel shifter used in multiplication in ARM. | 8 | L2 | 5 | 1 |
| b) | Explain the function of following instructions in ARM Microcontroller with an example.
i. MOV
ii. MVN
iii. MLA
iv. BIC
v. RSB | 8 | L2 | 5 | 1 |
| 8. a) | Explain any 4 Branch instruction of ARM with examples. | 8 | L2 | 5 | 1 |
| b) | Explain the features of ARM architecture and draw the register. | 8 | L2 | 5 | 1 |

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome.
